Homework 5 For Numeric II

1) Define $f_K = f(t_K, y_K)$.

Let $\varepsilon = \frac{t - t_K}{h}$, where $h = t_{K+1} - t_K$. Then Adams Batchforth says $y_{K+1} = y_K + \int_{t_K}^{t_{K+1}} f(t,y)\,dt$.

We need to construct Lagrange interpolation polynomial:

$$\Rightarrow p(t) = \sum_{J=1}^{m} f_{K+1-J} \prod_{\substack{i=1 \\ i\neq J}}^{m} \frac{t - t_{K+1-i}}{t_{K+1-J} - t_{K+1-i}}$$

$\Rightarrow p(\varepsilon) = f_K L_0(\varepsilon) + f_{K-1} L_1(\varepsilon) + f_{K-2} L_2(\varepsilon)$, our Lagrange basis are

$$\begin{cases} L_0(\varepsilon) = \frac{(\varepsilon+1)(\varepsilon+2)}{(0+1)(0+2)} = \frac{(\varepsilon+1)(\varepsilon+2)}{2} \\ L_1(\varepsilon) = \frac{(\varepsilon-0)(\varepsilon+2)}{(-1-0)(-1+2)} = \frac{\varepsilon(\varepsilon+2)}{-1} = -\varepsilon(\varepsilon+2) \\ L_2(\varepsilon) = \frac{\varepsilon(\varepsilon+1)}{(-2-0)(-2+1)} = \frac{\varepsilon(\varepsilon+1)}{2} \end{cases}$$

Combine the integral with the Lagrange interpolation polynomial, then

$$\varepsilon = \frac{t - t_K}{h} \Rightarrow t = t_K + h\varepsilon \Rightarrow dt = h\,d\varepsilon$$

It implies $\int_{t_K}^{t_{K+1}} p(t)\,dt = \int_0^1 p(t_K + h\varepsilon)\,h\,d\varepsilon = h\int_0^1 p(t_K + h\varepsilon)\,d\varepsilon$

when $t = t_K \Rightarrow \varepsilon = 0$

when $t = t_{K+1} = t_K + h \Rightarrow \varepsilon = 1$

thus, boundaries of integral has been changed.

f_K coefficient

$$\int_0^1 \frac{(\varepsilon+1)(\varepsilon+2)}{2}\, d\varepsilon = \frac{1}{2}\int_0^1 (\varepsilon^2+3\varepsilon+2)\, d\varepsilon$$

$$= \frac{1}{2}\left[\frac{\varepsilon^3}{3} + \frac{3\varepsilon^2}{2} + 2\varepsilon\right]_0^1 = \frac{1}{2}\cdot\frac{23}{6} = \frac{23}{12}$$

f_{K-1} coefficient

$$\int_0^1 -\varepsilon(\varepsilon+2)\, d\varepsilon = -\int_0^1 (\varepsilon^2+2\varepsilon)\, d\varepsilon = -\left[\frac{\varepsilon^3}{3} + \varepsilon^2\right]_0^1 = -\frac{4}{3} = -\frac{16}{12}$$

f_{K-2} coefficient

$$\int_0^1 \frac{\varepsilon(\varepsilon+1)}{2}\, d\varepsilon = \frac{1}{2}\int_0^1 (\varepsilon^2+\varepsilon)\, d\varepsilon = \frac{1}{2}\left[\frac{\varepsilon^3}{3} + \frac{\varepsilon^2}{2}\right]_0^1 = \frac{1}{2}\cdot\frac{5}{6} = \frac{5}{12}$$

Therefore,

$$y_{K+1} = y_K + h\left(\frac{23}{12} f_K - \frac{16}{12} f_{K-1} + \frac{5}{12} f_{K-2}\right).$$

$\square$